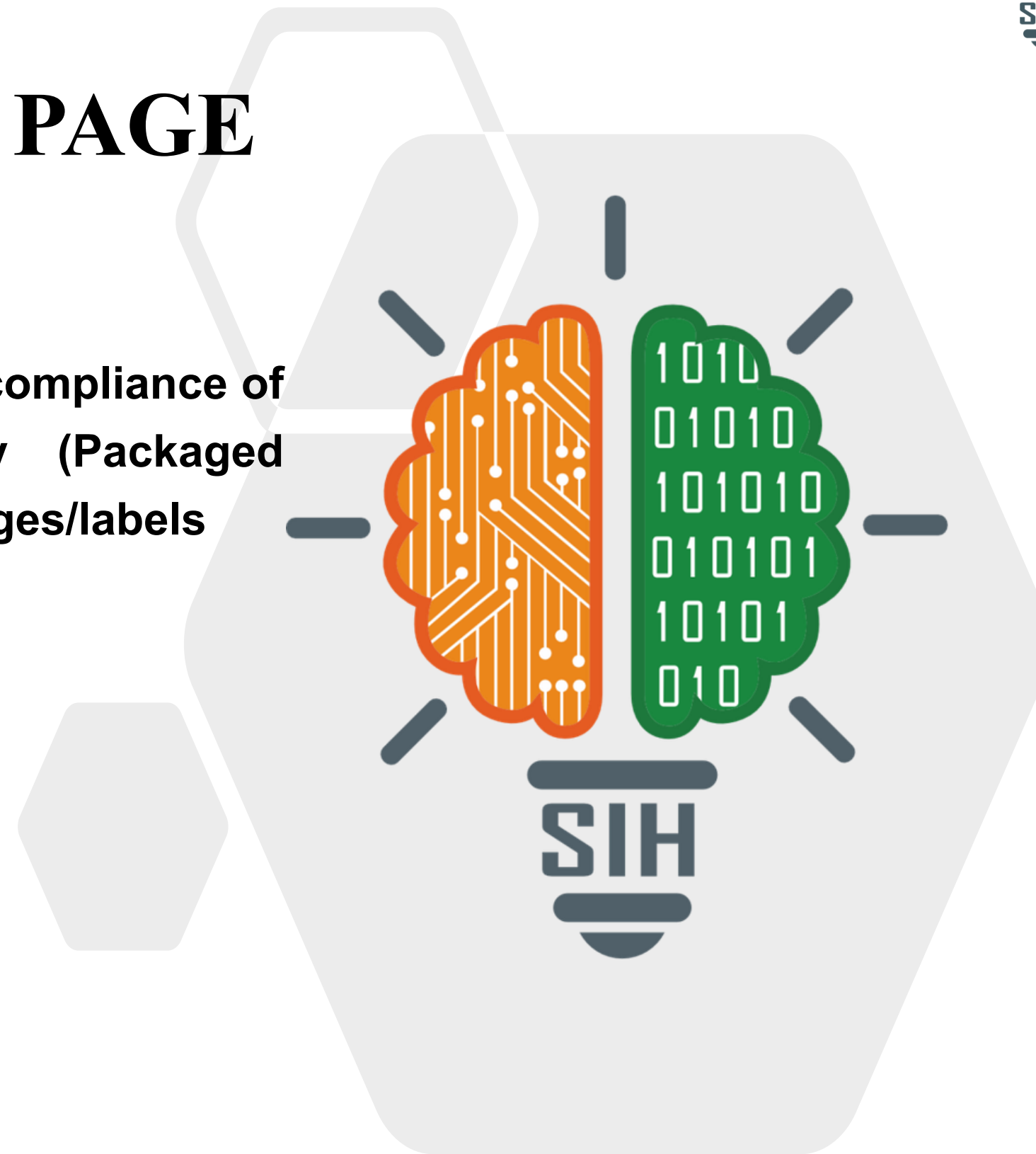


TITLE PAGE

- Problem Statement ID – SIH26034
- Problem Statement Title- Software System to check compliance of Packaged Commodities under Legal Metrology (Packaged Commodities) Rules, 2011 by scanning products/images/labels
- Theme- Miscellaneous
- PS Category- Software
- Team ID-
- Team Name- TEAM LUCIPHER





LabelSure – AI Compliance Scanner

SOLUTION:

- *Mobile-first app leverages Vision AI to analyze package photos, ensuring Legal Metrology compliance.*
- *Single-click feature generates legal notices, minimizing manual effort and swiftly addressing compliance issues.*
- *Intuitive interface allows quick photo capture and analysis, optimizing on-the-go efficiency.*
- *Designed to streamline compliance verification for officers with simplicity and effectiveness.*

ADDRESSING THE PROBLEM:

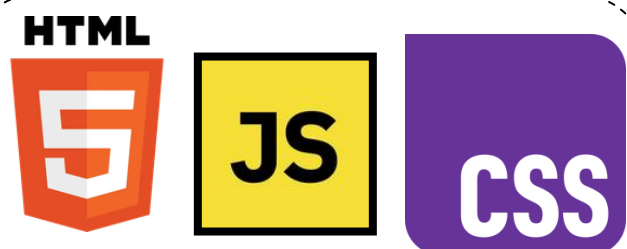
- *Faster Inspections: Cuts inspection time from 15–20 minutes per batch to much less, speeding up evaluations.*
- *Consistency and Reliability: Standardizes inspections, reducing human error for dependable results.*

INNOVATION & UNIQUENESS:

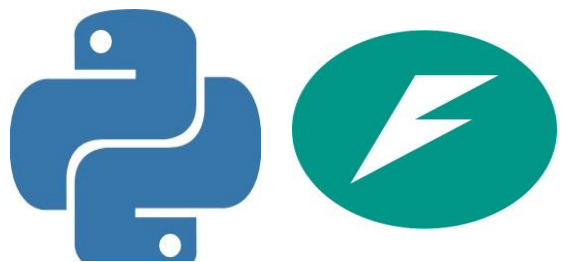
- *Integrates multimodal vision intelligence with Indian legal metrology constraints.*
- *Complies with LMPC Rules, 2011.*
- *Provides fast, near-instant feedback with automated compliance scores.*
- *Enables violation cross-checks on smartphones.*
- *Requires no native app installations.*

FRAMEWORKS:

FRONTEND



BACKEND



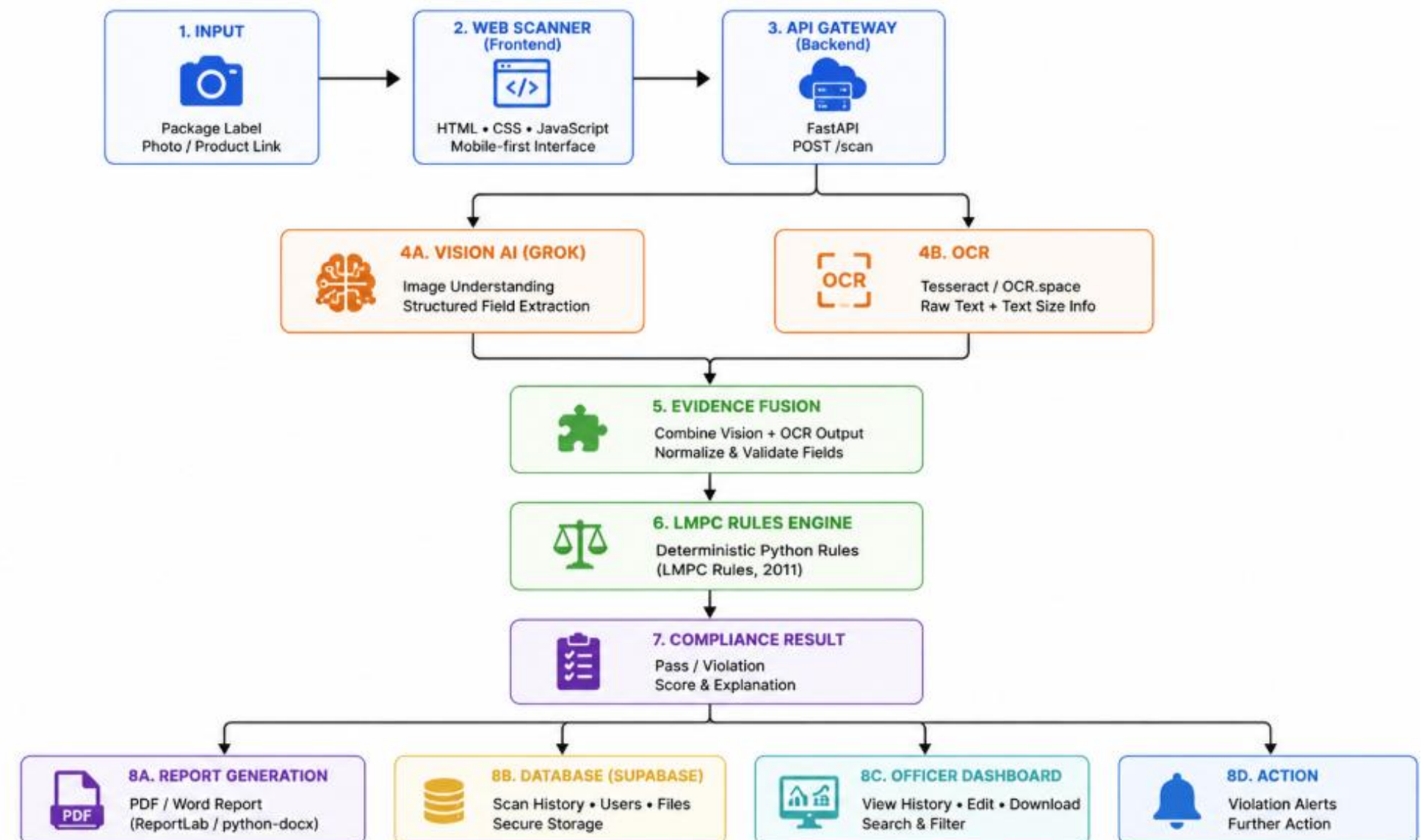
Frontend -> Html , Tailwind CSS ,
Javascript

Backend -> Python(FastAPI) + OCR

AI & VISION:



- groq API (vision model)
reads the label and
returns structured,
organized fields.
- OCR pass (Tesseract)
double-checks label text
and font size for fast,
near-real-time scoring.



ANALYSIS OF THE FEASIBILITY OF THE IDEA:

Our solution is technically feasible because it combines lightweight web technologies with a Python/FastAPI backend, Vision AI and OCR for extraction, and a rule-based compliance layer for verification. The modular architecture allows the AI, regulatory rules and reporting components to be updated independently without requiring specialized inspection hardware.

POTENTIAL CHALLENGES AND RISKS:

- *Image quality: Blur, glare, low lighting and angled packaging can reduce AI/OCR extraction accuracy.*
- *Label diversity: Different layouts, fonts, languages and package formats can make field extraction inconsistent.*
- *AI interpretation risk: Vision models can occasionally misread or omit information from complex labels.*

STRATEGIES FOR OVERCOMING CHALLENGES:

- *Image-quality checks + complementary Vision AI/OCR extraction reduce errors caused by poor photographs.*
- *Extracted fields are presented for officer review and correction before the final compliance result.*
- *Compliance is determined by explicit Python rules rather than relying on the AI model to make the final regulatory decision.*

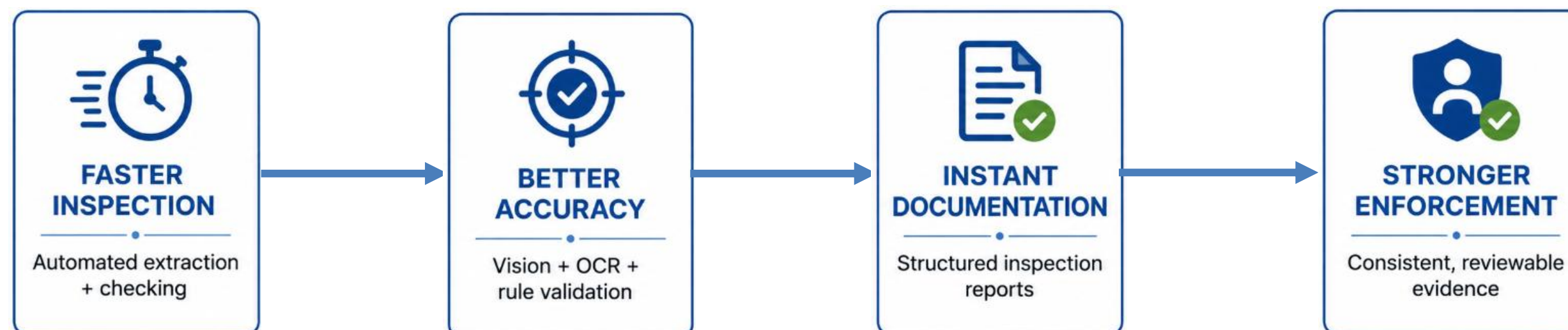
IMPACT AND BENEFITS:

POTENTIAL IMPACT ON TARGET AUDIENCE:

Legal Metrology Officers can inspect more products consistently by automating label-data extraction, compliance checks and report preparation, improving inspection coverage across retail and digital markets.

BENEFITS:

- *Faster inspections — automates repetitive label-data extraction and compliance checking.*
- *Better accuracy — combines Vision AI, OCR and deterministic rule validation to reduce missed declarations.*
- *Instant documentation — generates structured inspection reports without manually compiling findings.*
- *Stronger enforcement — gives officers consistent, reviewable compliance evidence across physical and digital markets.*





RESEARCH AND REFERENCES



Legal Metrology (Packaged Commodities) Rules, 2011 — the core regulatory framework used to verify mandatory declarations on packaged goods. Official source: Department of Consumer Affairs, Government of India.



Groq API — AI inference API used for package-label analysis and structured information extraction. Docs: console.groq.com/docs



Tesseract OCR — open-source OCR engine used as a secondary text-extraction and verification layer.



FastAPI — Python framework powering our backend API and inspection-processing workflow. Docs: fastapi.tiangolo.com



supabase — backend platform used for authentication, database storage, scan history, and file storage. Docs: supabase.com/docs